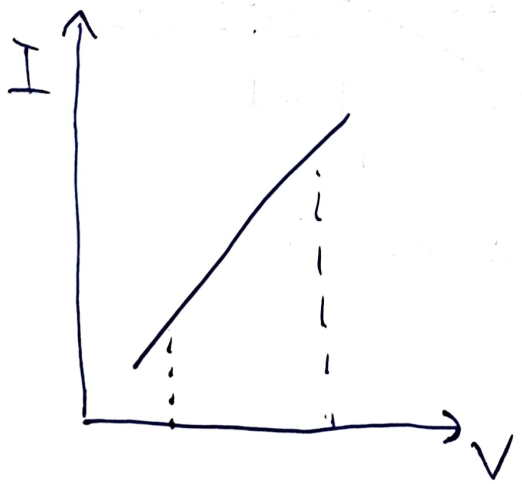
UNIT-1

1. **Ohms law:** At constant temp the current flowing through the conductor is $\propto$ to the voltage across it and inversely proportional to the resistance

$$\Rightarrow I \propto V$$
$$\Rightarrow I = \frac{V}{R}$$
$$\Rightarrow V = IR$$

limitations:

i) Resistivity changes with change in temp
ii) Relation b/w V & I depends upon sign of V
iii) Not applicable to semiconductors, transistors and diodes.
iv) It is not applicable with non linear elements of circuit
v) V and I ratio is not constant with time



2. KCL:

i, Total current passing towards a junction is equal to the total current passing away from the junction

ii, The algebraic sum of currents meeting at a junction is equal to 0

iii, It obeys Law of conservation of charge

iv, Figure ie $I_1 + I_2 = I_3 + I_4 + I_5$

KVL

i, The algebraic sum of EMFs and potential drops in a closed circuit is equal to 0

ii, The sum of all rising potential in equal to sum of all dropping potential.

iii, It obeys Law of conservation of energy

iv, Figure ie $V = V_1 + V_2 + V_3$

Resistance: $R = \frac{V}{I}$

Series: $R = R_1 + R_2 + \cdots$

parallel: $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$

Inductance:

$$V = L\frac{di}{dt}$$

Series: $L = L_1 + L_2 + \cdots$

parallel: $\frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2} + \cdots$

Capacitance

$$V = \frac{1}{C}\int I\,dt$$

Series: $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots$

parallel: $C_1 = C_1 + C_2 + \cdots$

Resistor problem Case 1: If R_1 & R_2 are in series then

Voltage across R_1 $V_1 = V\frac{R_1}{R_1 + R_2}$

Voltage across R_2 $V_2 = V\frac{R_2}{R_1 + R_2}$

Current is same in all positions

Case 2: If R_1 & R_2 are in parallel then

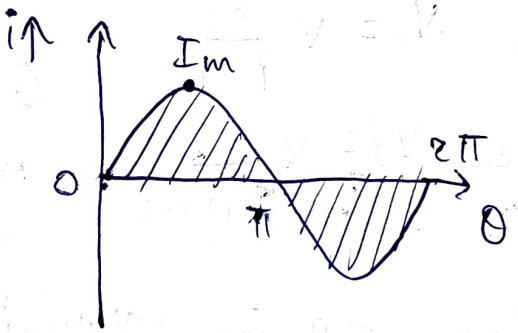
Current through R_1 $I_1 = I\frac{R_1}{R_1 + R_2}$

Current through R_2 $I_2 = I\frac{R}{R_1 + R_2}$

4. Define & Derive expressions for RMS, Average, Form factor & peak factor.

A. **RMS value:** The measure of the effective value of an alternating quantity. It is the square root of mean of squares of Instantaneous values of AC current

$$I_{Rms} = \sqrt{\frac{i_1^2 + i_2^2 + \cdots i_n^2}{n}}$$



For $I(\theta) = I_m \sin\theta$

limits = 0 to 2π

Time period $T = 2\pi$

$$I_{Rms} = \sqrt{\frac{1}{T}\int (I(\theta))^2 d\theta}$$

$$= \sqrt{\frac{1}{2\pi}\int_0^{2\pi} I_m^2 \sin^2\theta \, d\theta}$$

$$= \sqrt{\frac{I_m^2}{2\pi}\int_0^{2\pi}\left(\frac{1-\cos 2\theta}{2}\right) d\theta}$$

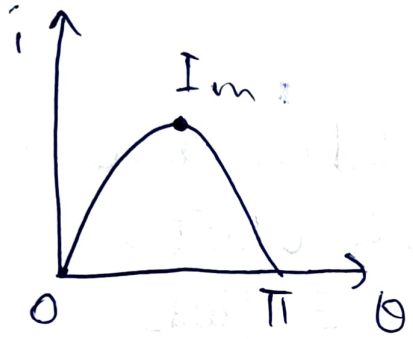
$$= \sqrt{\frac{I_m^2}{2\pi}\left[\frac{\theta}{2} - \frac{\sin 2\theta}{4}\right]_0^{2\pi}}$$

$$= \sqrt{\frac{I_m^2}{2\pi} \times \left[\frac{2\pi}{2} - 0\right]}$$

$$\boxed{I_{Rms} = \frac{I_m}{\sqrt{2}} = 0.707 I_m}$$

Average value: The Average of all instantaneous values taken over a half cycle

$$I_{AV} = \frac{i_1 + i_2 + \cdots i_n}{n}$$



$I(\theta) = I_m \sin\theta$ Limits: 0 to π

Time period $T = \pi$

$$I_{AV} = \frac{1}{T}\int I(\theta)\,d\theta$$

$$= \frac{1}{\pi}\int_0^{\pi} I_m \sin\theta\, d\theta$$

$$= \frac{I_m}{\pi}(-\cos\theta)_0^{\pi} = \frac{I_m}{\pi}(-\cos\pi + \cos 0) = \frac{I_m}{\pi}(1+1)$$

$$\boxed{I_{AV} = \frac{2}{\pi} I_m = 0.637\, I_m}$$

Peak Factor: Ratio of Max value of alternating quantity to RMS value of wave

$$\text{Peak Factor} = \frac{\text{Peak value}}{\text{RMS value}} = \frac{I_m}{I_m/\sqrt{2}} = \sqrt{2} \text{ (or) } 1.414$$

Form Factor: $\frac{\text{RMS}}{\text{Avg}} =$

$$= \frac{\frac{I_m}{\sqrt{2}}}{\frac{2I_m}{\pi}} = \frac{\pi}{2\sqrt{2}} = 1.110$$

9. Power: 3 Type

i, Active / True

→ Actual power consumed by circuit

→ Seen in heating a resistor / turning a motor

$$P = VI \cos\phi$$

ii, Reactive: $Q = VI \cos\phi$

→ Power that flows back and forth between source and load due to presence of inductors & capacitors

→ Used to maintain Mag-Fields & E-fields

iii, Apparent: $S = VI$

→ Total power flowing in a circuit

→ vector sum of Active and reactive power

Power Factor:

Ratio of Active power and apparent power

$$\text{Power factor} = \frac{\text{Active}}{\text{Apparent}} = \frac{P \cancel{I} \cos\phi}{\cancel{P I}} = \cos\phi$$

Importance:

i, Improves efficiency

ii, Maintains voltage levels at load ends

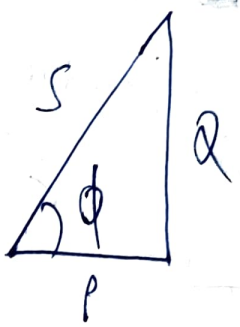
iii, High power factor reduces energy waste

6. Power △: Relation between active, reactive & apparent powers.

Active: def & Formula

Reactive: def & Formula

Apparent: def & Formula



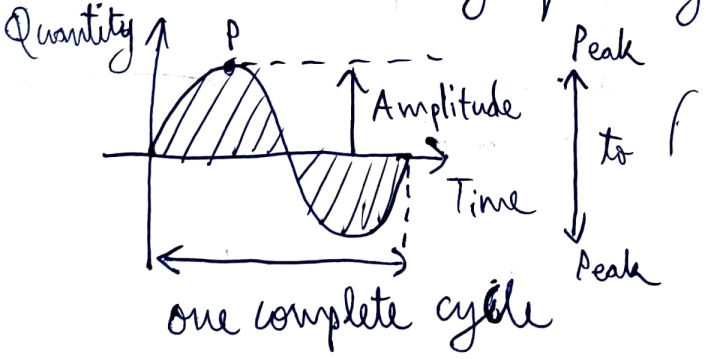
$$S = \sqrt{P^2 + Q^2}$$

$$S^2 = P^2 + Q^2$$

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7. Explain

i) Cycle: Complete set of +ve & -ve values of an alternating quantity



Time period: Time taken by quantity to complete one full cycle

Peak value: The Maximum Instantaneous value of the alternating quantity (P) (Draw graph)

Frequency: No of cycles per second of an alternating quantity

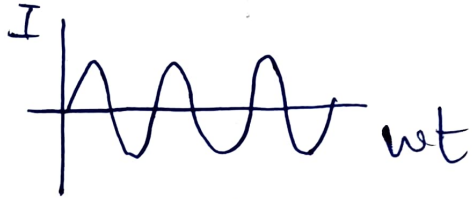
Peak-Peak value: Sum of Amplitudes of both half cycles (graph)

Amplitude: Magnitude of a half cycle of the Alternating quantity (graph)

8. AC

i, The Electric current which changes periodically with its direction & magnitude

ii, Graph



iii, Has two half cycles

iv, Bidirectional

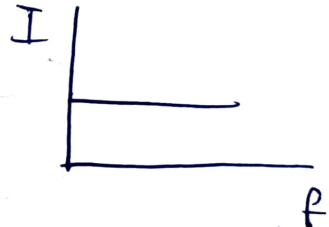
v, Frequency is defined = ω/t

vi, Time period is defined

DC

i, The steady flow of electric charge with constant magnitude

ii, Graph



iii, It is a direct flow

iv, Unidirectional

v, Frequency is 0

vi, Time period is ∞